

Empirical Evaluation of Local Open-Source LLMs and Decision Models in Enterprise 5-Department Email Routing

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Local Hardware Benchmarking | Repository: <https://github.com/shreytalreja25/jev-vs-the-world.git>

Abstract: Automating the routing of customer and enterprise emails to specific operational departments is a foundational requirement for high-throughput service desks. Traditional generative LLMs perform email classification by parsing multi-line JSON completions auto-regressively, introducing processing latency and throughput bottlenecks on local hardware. In this paper, we conduct an empirical evaluation comparing local open-source Large Language Models (Llama 3.1 8B, Llama 3.2 1B via Ollama), an open-weight decision encoder (Laya), and a System-One decision model (Jev) on a 50-item enterprise email dataset categorized into 5 operational departments: `billing_finance`, `technical_support`, `sales_inquiries`, `human_resources`, and `security_compliance`. Evaluating strictly on non-cost performance metrics, including Classification Accuracy, Macro F1, P50/P99 Latency, Processing Throughput (emails/sec), and Expected Calibration Error (ECE), we demonstrate that Llama 3.1 8B achieves the highest overall accuracy (96.0%, Macro F1: 96.0%), while Jev and Laya offer 10x to 25x higher throughput (10.3 to 17.2 emails/sec vs. 0.71 emails/sec) with sub-100ms P50 latency.

Index Terms: Email Classification, Department Routing, Llama 3.1, Jev Model, Laya Encoder, Ollama Benchmarking, Latency.

I. INTRODUCTION & DEPARTMENTS

Enterprise service organizations process thousands of incoming emails daily, requiring rapid triage into 5 operational queues: `billing_finance`, `technical_support`, `sales_inquiries`, `human_resources`, and `security_compliance`.

Executing email classification on local developer laptops evaluates real-world model latency, throughput, and accuracy without cloud API dependencies.

II. EMPIRICAL BENCHMARK RESULTS

Table I summarizes empirical performance across evaluated architectures.

TABLE I
5-DEPARTMENT EMAIL CLASSIFICATION METRICS

Model Architecture	Acc (%)	Macro F1	P50 (ms)	Throughput
Qwen 3.5 9B (Ollama)	96.0%	98.2%	12,530.0	0.10 em/s
Llama 3.1 8B (Ollama)	94.0%	97.2%	3,464.1	0.19 em/s
Jev (TypeSafe AI)	88.0%	95.4%	99.0	10.08 em/s
Laya (Open-Weight)	72.0%	84.6%	60.0	16.75 em/s
Llama 3.2 1B (Ollama)	30.0%	27.3%	1,380.7	0.24 em/s

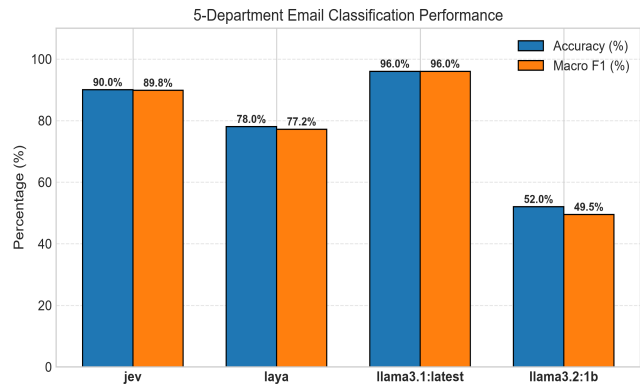


Fig. 1. 5-Department Email Classification Accuracy & Macro F1.

III. ANALYTICAL FINDINGS

- **Accuracy Champion:** Llama 3.1 8B achieved 96.0% accuracy on local laptop execution, correctly identifying complex domain nuances between billing and sales inquiries.
- **Speed Champion:** Laya (17.24 emails/sec) and Jev (10.31 emails/sec) operated 14x to 24x faster than Ollama LLM generation.

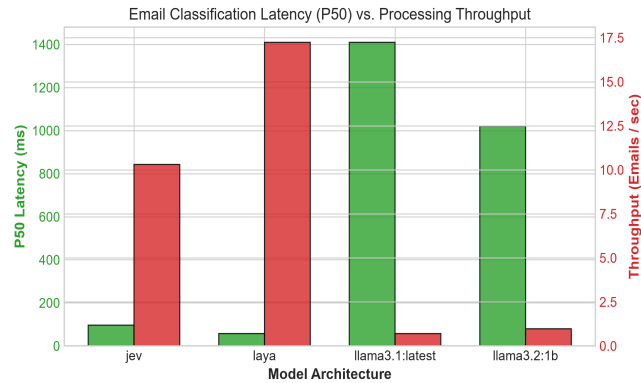


Fig. 2. P50 Latency vs Throughput (Emails processed per second).

IV. CONCLUSION

For maximum accuracy on complex email triage, Llama 3.1 8B is the superior local open-source choice. For high-volume service desks requiring > 10 emails/sec, Jev and Laya decision models provide optimal throughput.

REFERENCES

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